

tf.compat.v1.constant_initializer Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/compat/v1/constant_initializer

Initializer that generates tensors with constant values.

Function Signatures

```
tf.compat.v1.constant_initializer( value=0,  
dtype=tf.dtypes.float32, verify_shape=False )
```

Code Examples

```
tf.compat.v1.constant_initializer(  
value=0,  
dtype=tf.dtypes.float32,  
verify_shape=False  
)
```

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value=0,  
dtype=tf.dtypes.float32,  
verify_shape=False  
)
```

```
tf.dtypes.float32
```

```
value = [0, 1, 2, 3, 4, 5, 6, 7]
initializer = tf.compat.v1.constant_initializer(
    value=value,
    dtype=tf.float32,
    verify_shape=False)
variable = tf.Variable(initializer(shape=[2, 4]))
```

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value = [0, 1, 2, 3, 4, 5, 6, 7]
initializer = tf.compat.v1.constant_initializer(
    value=value,
    dtype=tf.float32,
    verify_shape=False)
variable = tf.Variable(initializer(shape=[2, 4]))
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value = [0, 1, 2, 3, 4, 5, 6, 7]
initializer = tf.constant_initializer(value=value)
tf.Variable(initializer(shape=[2, 4], dtype=tf.float32))
```

```
value = [0, 1, 2, 3, 4, 5, 6, 7]
initializer = tf.constant_initializer(value=value)
tf.Variable(initializer(shape=[2, 4], dtype=tf.float32))
```

```
value = [1., 2., 3., 4.]
initializer = tf.compat.v1.constant_initializer(
    value=value, dtype=tf.float32, verify_shape=True)
tf.Variable(initializer(shape=[2, 2])).numpy()
Traceback (most recent call last):
TypeError: Expected Tensor's shape: (2, 2), got (4,).
initializer = tf.compat.v1.constant_initializer(
    value=value, dtype=tf.float32, verify_shape=False)
tf.Variable(initializer(shape=[2, 2])).numpy()
array([[1., 2.],
       [3., 4.]], dtype=float32)
```

Documentation

Initializer that generates tensors with constant values.

View aliases

Compat aliases for migration

See

Migration guide for
more details.

`tf.compat.v1.initializers.constant`

Migrate to TF2

Although it is a legacy API endpoint, `tf.compat.v1.constant_initializer` is compatible with eager execution and `tf.function`.

To migrate to a non-legacy TF2 API, please use `tf.constant_initializer` instead. The `dtype` argument in `tf.compat.v1.constant_initializer.init_()` does not exist in `tf.constant_initializer.init_()`. However, you can specify the `dtype` in `__call__()` in both cases.

In the `compat.v1` symbol, if `verify_shape` is set to `True`, an exception is raised when initializing a variable with a different shape from `value`. If set to `False`, `value` is reshaped to initialize the variable if necessary. An exception would only be raised when the number of elements are different.

The `verify_shape` argument is not supported in TF2. Using

`tf.constant_initializer` is equivalent to setting `verify_shape` to `False`.

Structural Mapping to TF2

Before:

After:

How to Map Arguments

Before & After Usage Example

Before:

After:

Description

Used in the notebooks

- Migrating model checkpoints
- Use TF1.x models in TF2 workflows

The resulting tensor is populated with values of type `dtype`, as specified by arguments `value` following the desired shape of the new tensor (see examples below).

The argument `value` can be a constant value, or a list of values of type `dtype`. If `value` is a list, then the length of the list must be less than or equal to the number of elements implied by the desired shape of the tensor. In the case where the total number of elements in `value` is less

than the number of elements required by the tensor shape, the last element in value will be used to fill the remaining entries. If the total number of elements in value is greater than the number of elements required by the tensor shape, the initializer will raise a `ValueError`.

Raises

Examples

Methods

from_config

[View source](#)

Instantiates an initializer from a configuration dictionary.

Example:

get_config

[View source](#)

Returns the configuration of the initializer as a JSON-serializable dict.

`__call__`

[View source](#)

Returns a tensor object initialized as specified by the initializer.